

WASP-39b

R-1273

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-39_220327 · created 2026-07-31T14:08:00+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459665.8949 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1912 +/- 0.0048
Transit depth (Rp/Rs)^2	3.65 +/- 0.18 [%]
Orbital Inclination (inc)	88.04 +/- 0.74
Airmass coefficient 1 (a1)	1.077 +/- 0.016
Airmass coefficient 2 (a2)	-0.057 +/- 0.038
Scatter in the residuals of the lightcurve fit is	1.51 %
Best Comparison Star	#1 - [434, 220]
Optimal Aperture	3.48
Optimal Annulus	11.29
Transit Duration (day)	0.0917 +/- 0.0035

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1912 ± 0.0048 vs 0.1457 (deviation 6.58σ)
Duration fit vs published	0.0917 d vs 0.12 d (deviation 5.62σ)
Mid-time O–C	-5.0 min
Depth SNR	27.7
Residual scatter	1.51 %

Light curve

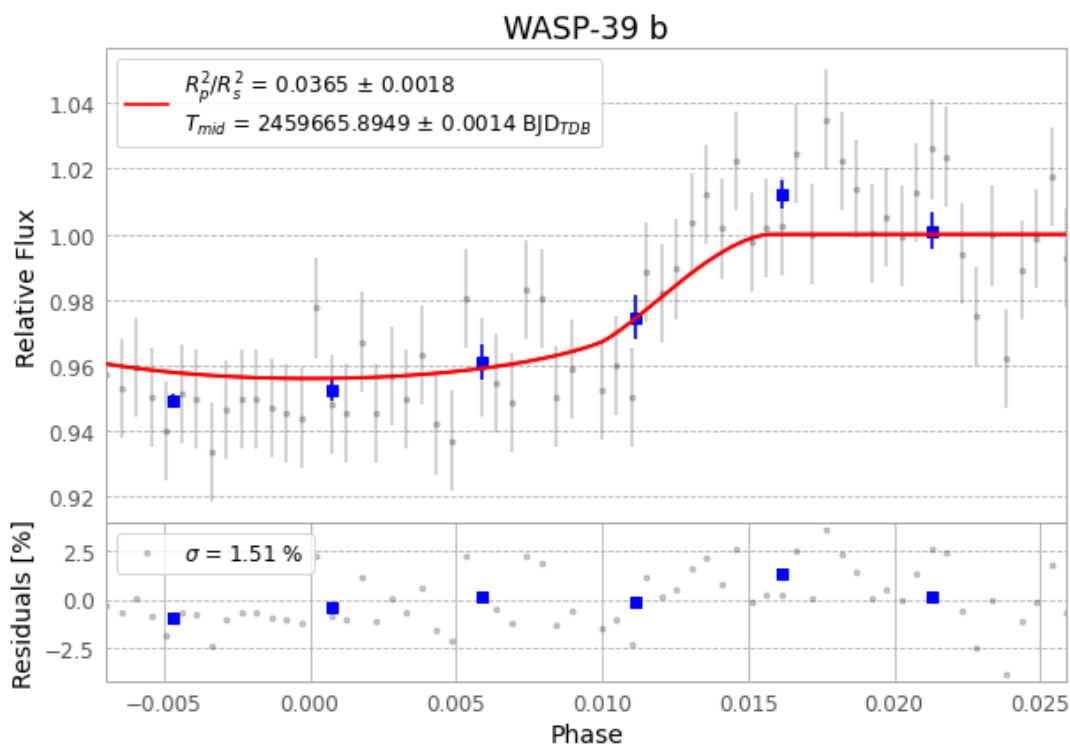


Figure 1 — FinalLightCurve_WASP-39 b_2022-03-27.png (SHA-256 19ebe0d5350d2369...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [192, 231], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 4d111645769450db...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

65 FITS frames (43 MB), WASP-39220327083913.FITS → WASP-39220327115113.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-39-220327.log (3ecd482e3b37...), FinalLightCurve_WASP-39 b_2022-03-27.png (19ebe0d5350d...), FinalLightCurve_WASP-39 b_2022-03-27.csv (a51b4339c0bf...)

Compute resources

Wall time 225.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 14:08Z.